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Cooper

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(54) **TENSIONED SUPPORT POST AND OTHER
MOLTEN METAL DEVICES**

(58) **Field of Classification Search**

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F05B 2240/90

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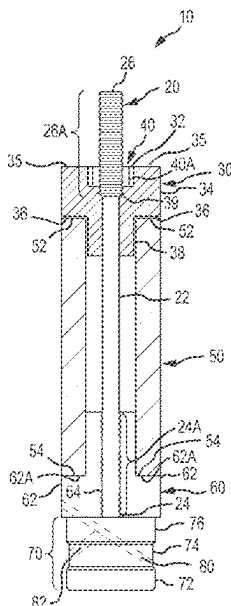
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(57)

ABSTRACT

A vertically-elongated member, which is preferably a sup-
port post used in a molten metal pump, includes a ceramic
tube and tensioning structures to add a compressive load to
the tube along its longitudinal axis. This makes the tube less
prone to breakage. Another vertically-elongated member,
such as a support post, includes one or more reinforcement
members to help alleviate breakage. A device, such as a
pump, used in a molten metal bath includes one or more of
such vertical members.

20 Claims, 11 Drawing Sheets



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continuation of application No. 17/496,229, filed on Oct. 7, 2021, now Pat. No. 11,976,672, which is a continuation of application No. 16/195,678, filed on Nov. 19, 2018, now Pat. No. 11,149,747.

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(58) **Field of Classification Search**

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See application file for complete search history.

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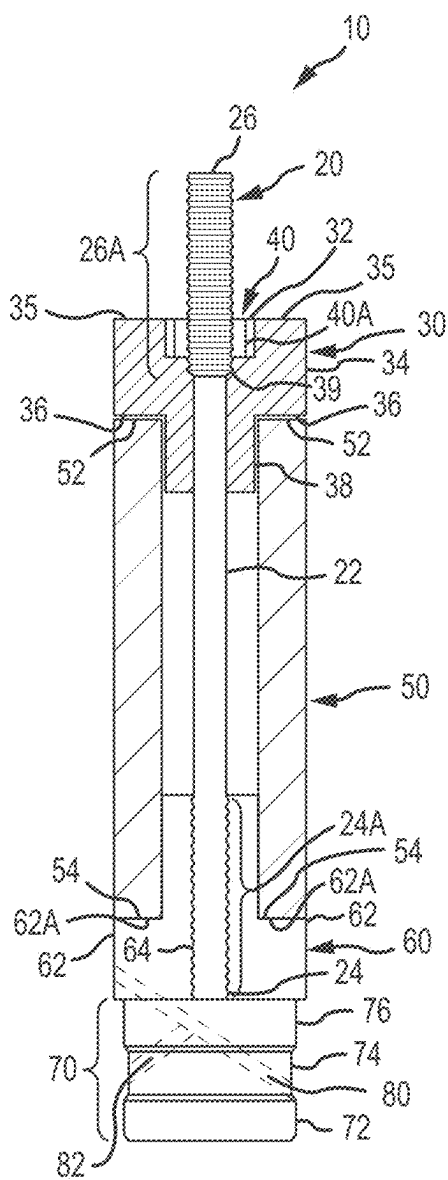


FIG.1

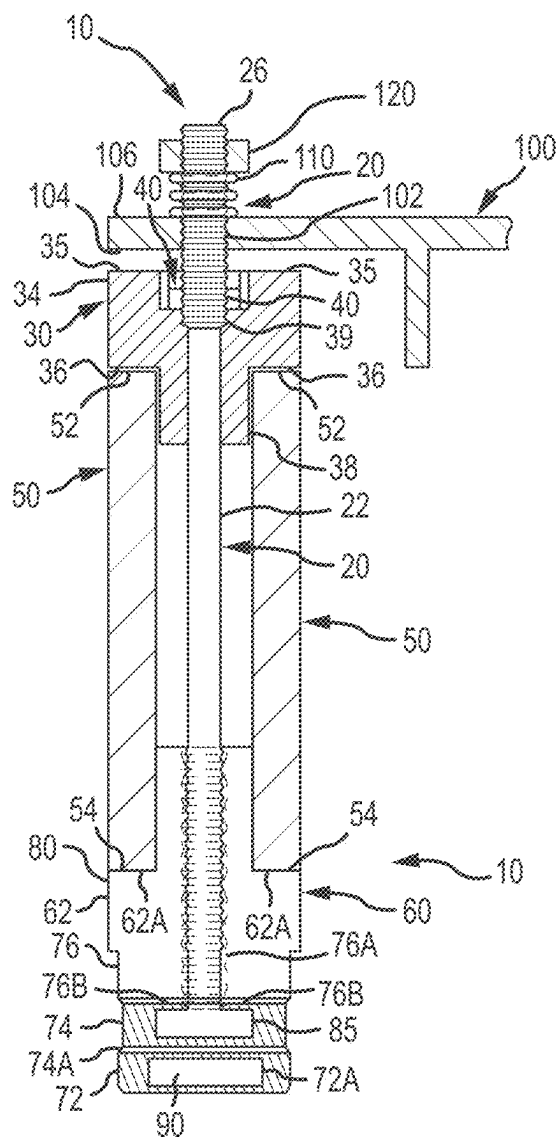


FIG.2

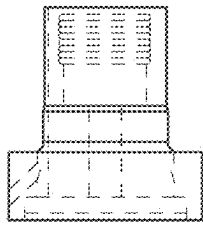


FIG. 2B

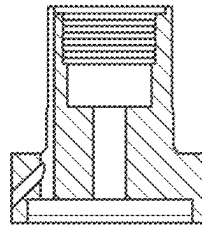


FIG. 2E

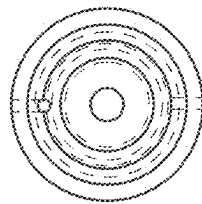


FIG. 2C

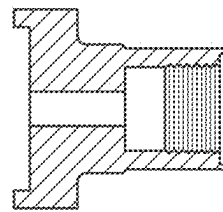
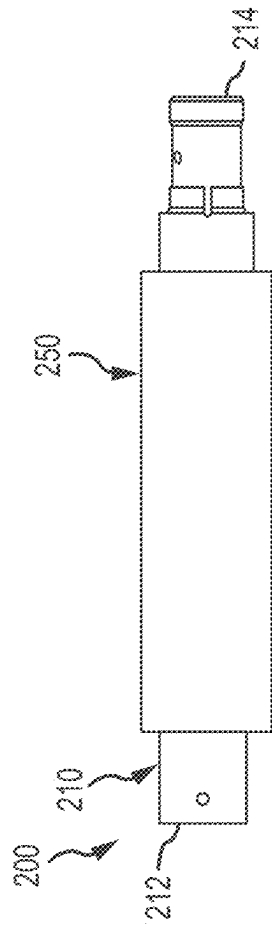
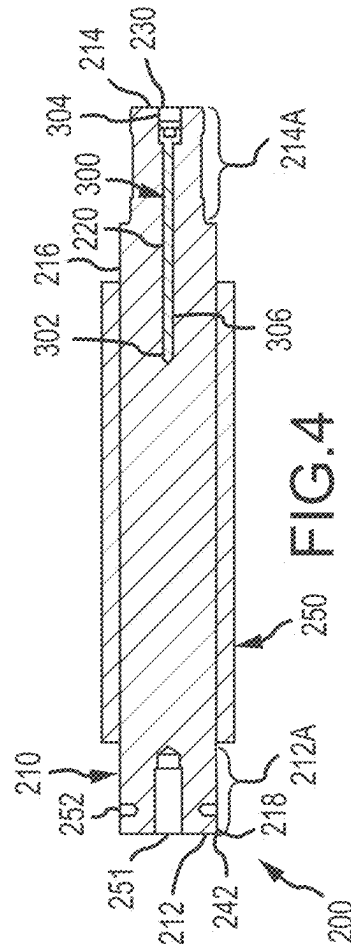


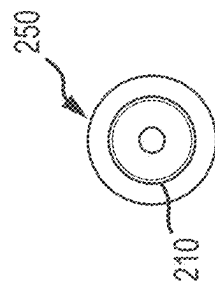
FIG. 2D



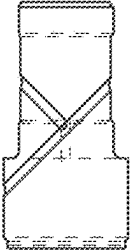
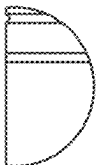
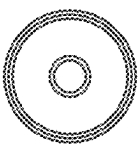
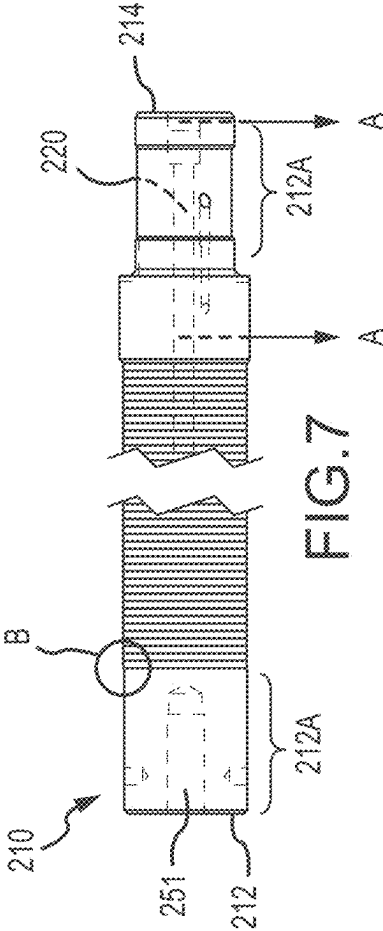
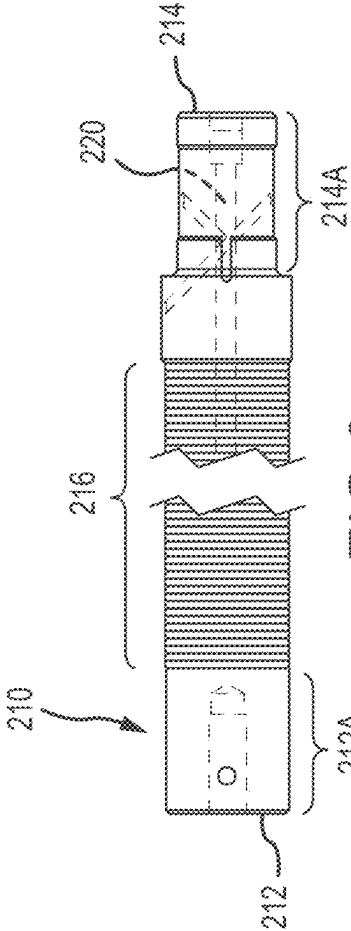
30



4
G
L







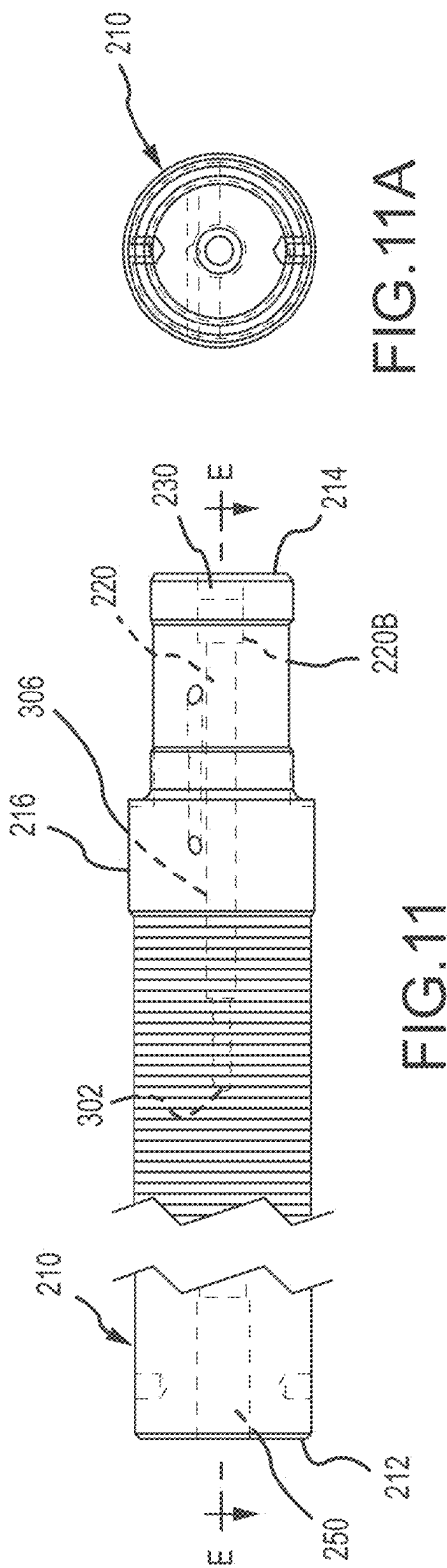


FIG. 11A

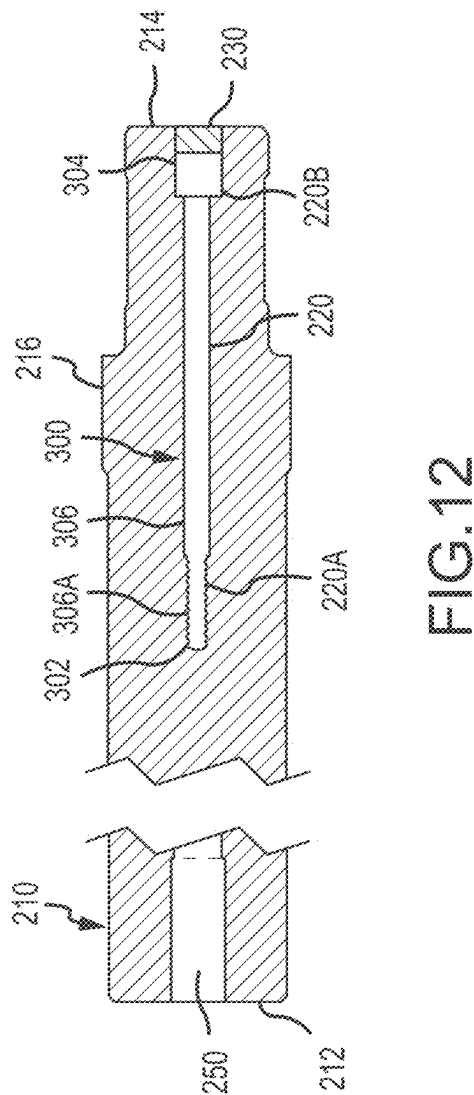


FIG. 12



FIG. 13

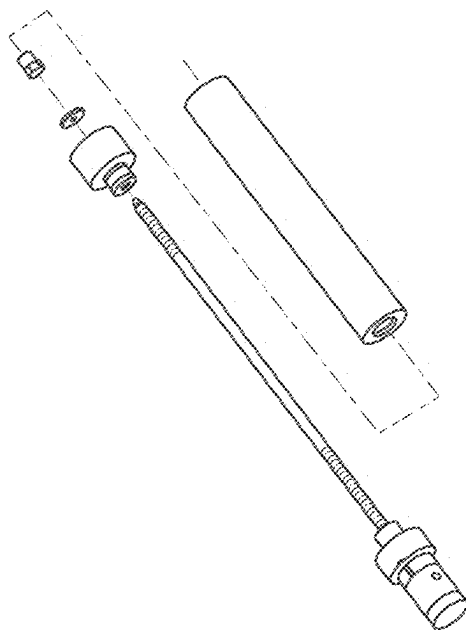


FIG. 14

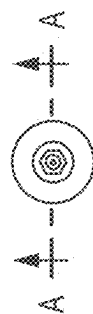


FIG. 15

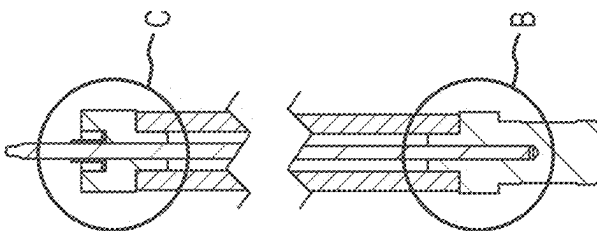


FIG. 16

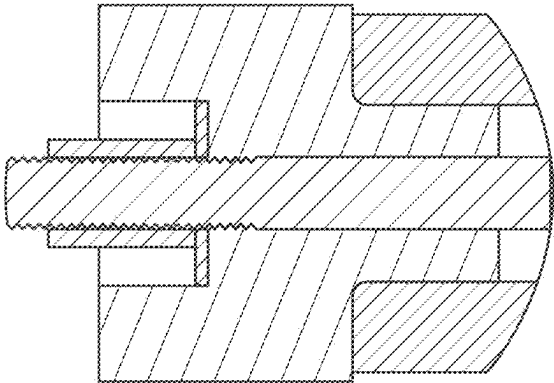


FIG.18

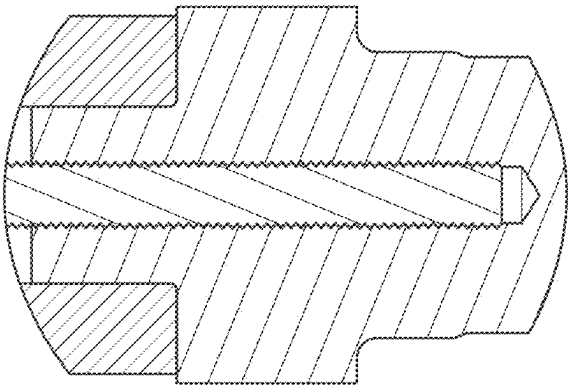


FIG.17

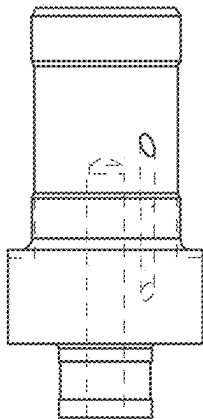


FIG.19

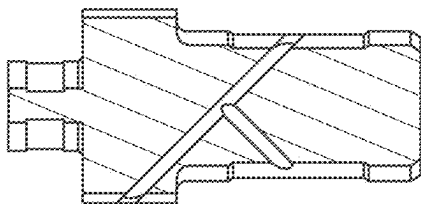


FIG.22

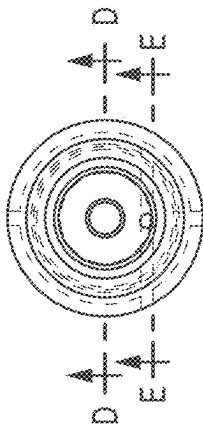


FIG.20

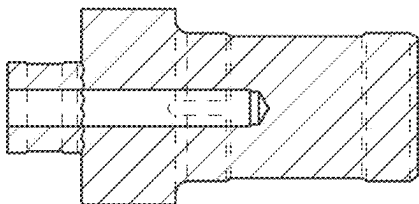


FIG.21

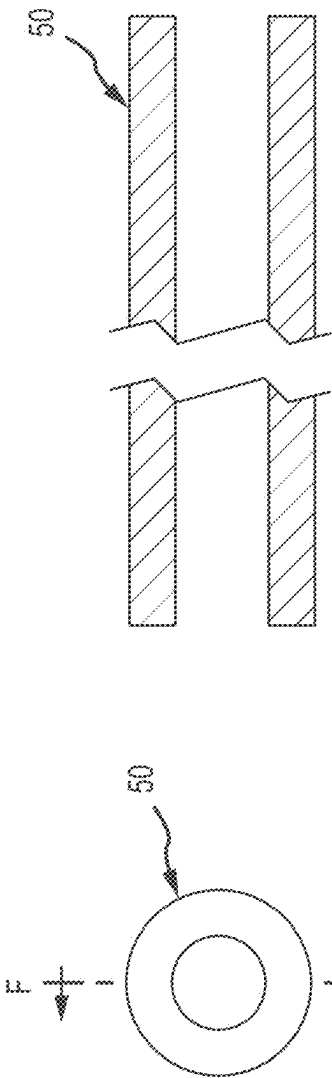


FIG. 23

FIG. 25

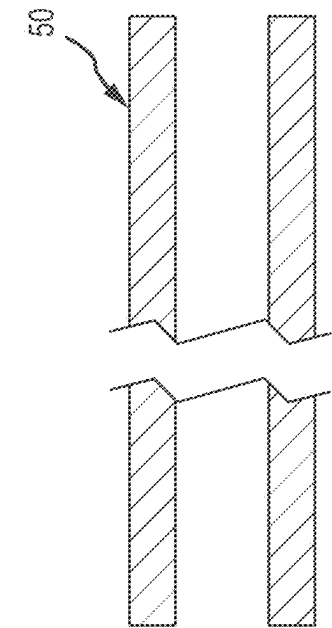
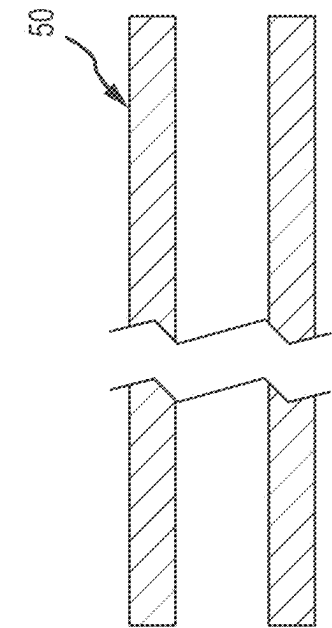


FIG. 24



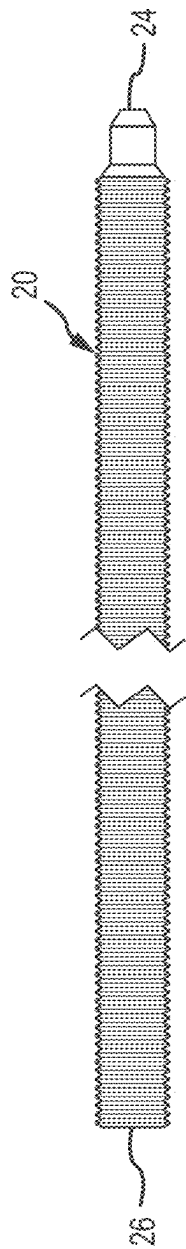


FIG.27

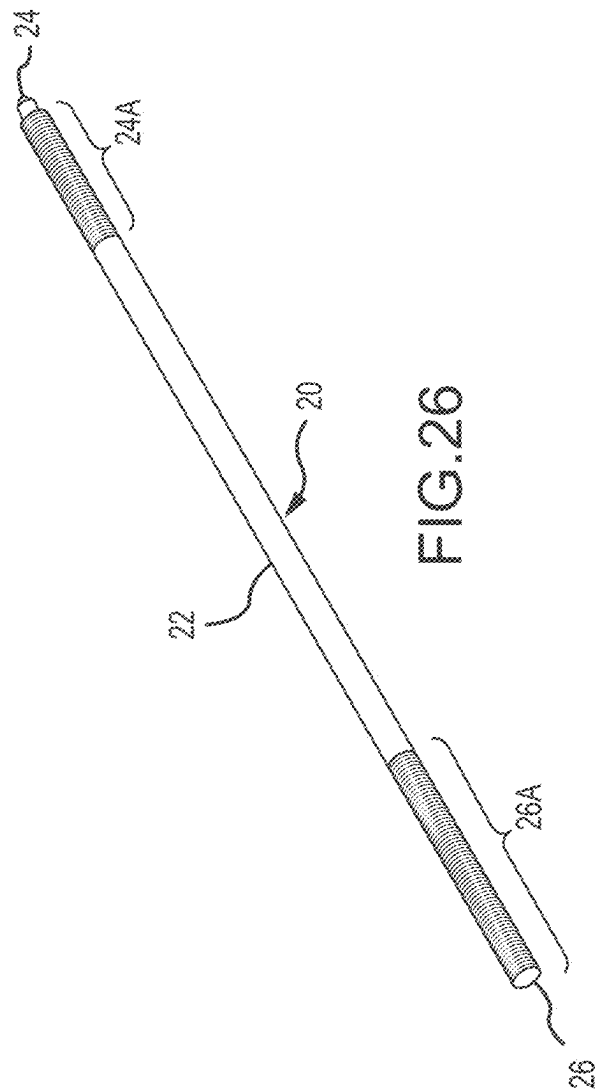


FIG.26

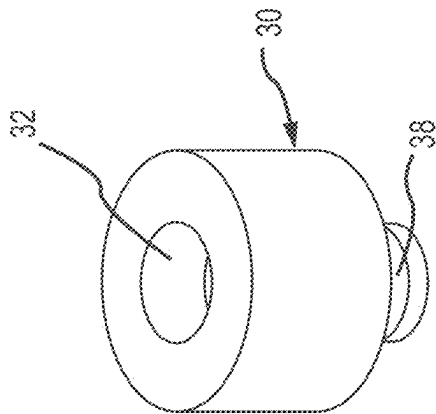


FIG. 28

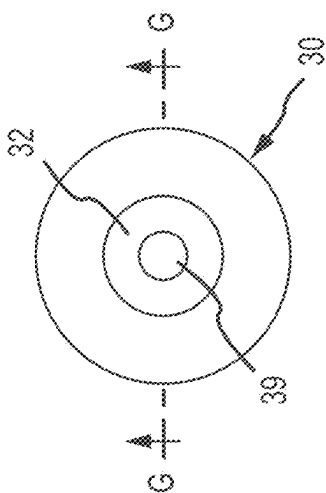


FIG. 29

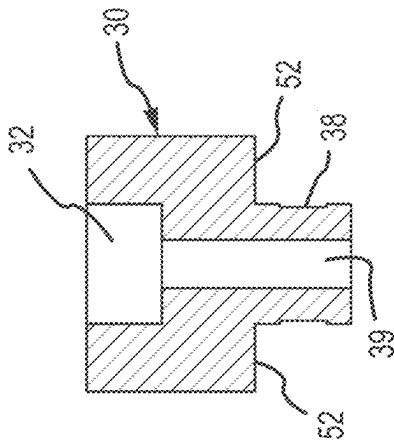


FIG. 30

TENSIONED SUPPORT POST AND OTHER MOLTEN METAL DEVICES

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a continuation of, and claims priority to U.S. patent application Ser. No. 18/139,936, filed Apr. 26, 2023 and entitled “Tensioned Support Post and Other Molten Metal Devices,” which is a continuation of, and claims priority to U.S. patent application Ser. No. 17/496,229, filed Oct. 7, 2021 and entitled “Tensioned Support Post and Other Molten Metal Devices,” which is a continuation of, and claims priority to U.S. patent application Ser. No. 16/195,678 (Now U.S. Pat. No. 11,149,747), filed Nov. 19, 2018, and entitled “Tensioned Support Posts and Other Molten Metal Devices” which claims priority to U.S. Provisional Application 62/588,090, filed Nov. 17, 2017, and entitled “Tensioned Support Post and Other Molten Metal Devices,” each of the disclosures of which are incorporated herein by reference. This Application incorporates by reference U.S. application Ser. No. 15/406,515 (now U.S. Pat. No. 10,267,314), filed Jan. 13, 2017, and entitled “Tensioned Support Shaft and Other Molten Metal Devices,” to the extent such application does not conflict with the present disclosure.

FIELD

The invention relates to tensioned support posts and other components, such as a reinforced support post that may be used in pumps for pumping molten metal.

BACKGROUND

As used herein, the term “molten metal” means any metal or combination of metals in liquid form, such as aluminum, copper, iron, zinc, and alloys thereof. The term “gas” means any gas or combination of gases, including argon, nitrogen, chlorine, fluorine, Freon, and helium, which are released into molten metal.

Known molten-metal pumps include (a) a pump base (also called a housing or casing), (b) one or more inlets (an inlet being an opening in the housing to allow molten metal to enter a pump chamber), (c) a pump chamber of any suitable configuration, which is an open area formed within the housing, (d) a discharge, which is a channel or conduit of any structure or type communicating with the pump chamber (in an axial pump the chamber and discharge may be the same structure or different areas of the same structure) and that leads from the pump chamber to (e) an outlet, which is an opening formed in the exterior of the housing through which molten metal exits the casing. An impeller, also called a rotor, is mounted at least partially in the pump chamber and is connected to a drive system. The drive shaft is typically (a) an impeller shaft having one end connected to the impeller and the other end connected to a coupling, and (b) a motor shaft having one end connected to a motor (such as an electric, hydraulic, or pneumatic motor) and the other end connected to the coupling. Often, the impeller (or rotor) shaft is comprised of graphite and/or ceramic (such as silicon carbide), the motor shaft is comprised of steel, and the coupling is comprised of steel.

As the motor turns the drive shaft, the drive shaft turns the impeller and the impeller pushes molten metal out of the pump chamber, through the discharge, out of the outlet and into the molten metal bath. Most molten metal pumps are gravity fed, wherein gravity forces molten metal through the

inlet and into the pump chamber as the impeller pushes molten metal out of the pump chamber.

Some molten metal pumps do not include a base or support posts and are sized to fit into a structure by which molten metal is pumped. Most pumps have a metal platform, or superstructure, that is either supported by a plurality of support posts attached to the pump base, or supported by another structure if there is no pump base. The motor is positioned on the superstructure, if a superstructure is used.

This application incorporates by reference the portions of the following documents that are not inconsistent with this disclosure: U.S. Pat. No. 4,598,899, issued Jul. 8, 1986, to Paul V. Cooper, U.S. Pat. No. 5,203,681, issued Apr. 20, 1993, to Paul V. Cooper, U.S. Pat. No. 5,308,045, issued May 3, 1994, to Paul V. Cooper, U.S. Pat. No. 5,662,725, issued Sep. 2, 1997, to Paul V. Cooper, U.S. Pat. No. 5,678,807, issued Oct. 21, 1997, to Paul V. Cooper, U.S. Pat. No. 6,027,685, issued Feb. 22, 2000, to Paul V. Cooper, U.S. Pat. No. 6,124,523, issued Sep. 26, 2000, to Paul V. Cooper, U.S. Pat. No. 6,303,074, issued Oct. 16, 2001, to Paul V. Cooper, U.S. Pat. No. 6,689,310, issued Feb. 10, 2004, to Paul V. Cooper, U.S. Pat. No. 6,723,276, issued Apr. 20, 2004, to Paul V. Cooper, U.S. Pat. No. 7,402,276, issued Jul. 22, 2008, to Paul V. Cooper, U.S. Pat. No. 7,470,392, issued Dec. 30, 2008, to Paul V. Cooper, U.S. Pat. No. 7,507,367, issued Mar. 24, 2009, to Paul V. Cooper, U.S. Pat. No. 7,906,068, issued Mar. 15, 2011, to Paul V. Cooper, U.S. Pat. No. 8,075,837, issued Dec. 13, 2011, to Paul V. Cooper, U.S. Pat. No. 8,110,141, issued Feb. 7, 2012, to Paul V. Cooper, U.S. Pat. No. 8,178,037, issued May 15, 2012, to Paul V. Cooper, U.S. Pat. No. 8,337,746, issued Dec. 25, 2012, to Paul V. Cooper, U.S. Pat. No. 8,361,379, issued Jan. 29, 2013, to Paul V. Cooper, U.S. Pat. No. 8,366,993, issued Feb. 5, 2013, to Paul V. Cooper, U.S. Pat. No. 8,409,495, issued Apr. 2, 2013, to Paul V. Cooper, U.S. Pat. No. 8,440,135, issued May 15, 2013, to Paul V. Cooper, U.S. Pat. No. 8,444,911, issued May 21, 2013, to Paul V. Cooper, U.S. Pat. No. 8,449,814, issued May 28, 2013, to Paul V. Cooper, U.S. Pat. No. 8,475,708, issued Jul. 2, 2013, to Paul V. Cooper, U.S. Pat. No. 8,501,084, issued Aug. 6, 2013, to Paul V. Cooper, U.S. patent application Ser. No. 12/895,796, filed Sep. 30, 2010, to Paul V. Cooper, U.S. patent application Ser. No. 13/791,952, filed Mar. 9, 2013, to Paul V. Cooper, U.S. Pat. No. 8,529,828, issued Sep. 10, 2013, to Paul V. Cooper, U.S. Pat. No. 8,535,603 issued Sep. 17, 2013, to Paul V. Cooper, U.S. Pat. No. 8,613,884, issued Dec. 24, 2013 to Paul V. Cooper, U.S. Pat. No. 8,714,914, issued May 6, 2014 to Paul V. Cooper, U.S. Pat. No. 8,753,563, issued Jun. 17, 2014, to Paul V. Cooper, U.S. Pat. No. 9,011,761, issued Apr. 21, 2015, to Paul V. Cooper, U.S. Pat. No. 9,017,597, issued Apr. 28, 2015, to Paul V. Cooper, U.S. Pat. No. 9,034,244, issued May 19, 2015, to Paul V. Cooper, U.S. Pat. No. 9,080,577, issued Jul. 14, 2015, to Paul V. Cooper, U.S. Pat. No. 9,108,244, issued Aug. 18, 2015, to Paul V. Cooper, U.S. Pat. No. 9,156,087, issued Oct. 13, 2015, to Paul V. Cooper, U.S. Pat. No. 9,205,490, issued Dec. 8, 2015, to Paul V. Cooper, U.S. Pat. No. 9,328,615, issued May 3, 2016 to Paul V. Cooper, U.S. Pat. No. 9,377,028, issued Jun. 28, 2016, to Paul V. Cooper, U.S. Pat. No. 9,382,599, issued Jul. 5, 2016, to Paul V. Cooper, U.S. Pat. No. 9,383,140, issued Jul. 5, 2016, to Paul V. Cooper, U.S. Pat. No. 9,409,232, issued Aug. 9, 2016, to Paul V. Cooper, U.S. Pat. No. 9,410,744, issued Aug. 9, 2016, to Paul V. Cooper, U.S. Pat. No. 9,422,942, issued Aug. 23, 2016, to Paul V. Cooper, U.S. Pat. No. 9,435,343, issued Sep. 6, 2016, to Paul V. Cooper, U.S. Pat. No. 9,464,636, issued Oct. 11, 2016, to Paul V. Cooper, U.S. Pat. No.

9,470,239, issued Oct. 18, 2016, to Paul V. Cooper, 9,481, 035, issued Nov. 1, 2016, to Paul V. Cooper, U.S. Pat. No. 9,482,469, issued Nov. 1, 2016, to Paul V. Cooper, U.S. Pat. No. 9,506,129, issued Nov. 29, 2016, to Paul V. Cooper, U.S. Pat. No. 9,566,645, issued Feb. 14, 2017, to Paul V. Cooper, 9,581,388, issued Feb. 28, 2017, to Paul V. Cooper, U.S. Pat. No. 9,587,883, issued Mar. 7, 2017, to Paul V. Cooper, U.S. Pat. No. 9,643,247, issued May 9, 2017, to Paul V. Cooper, U.S. Pat. No. 9,657,578, issued May 23, 2017, to Paul V. Cooper, U.S. Pat. No. 9,855,600, issued Jan. 2, 2018, to Paul V. Cooper, U.S. Pat. No. 9,862,026, issued Jan. 9, 2018, to Paul V. Cooper, U.S. Pat. No. 9,903,383, issued Feb. 27, 2018, to Paul V. Cooper, U.S. Pat. No. 9,909,808, issued Mar. 6, 2018, to Paul V. Cooper, U.S. Pat. No. 9,925,587, issued Mar. 27, 2018, to Paul V. Cooper, entitled U.S. Pat. No. 9,982,945, issued May 29, 2018, to Paul V. Cooper, U.S. Pat. No. 10,052,688, issued Aug. 21, 2018, to Paul V. Cooper, U.S. Pat. No. 10,072,891, issued Sep. 11, 2018, to Paul V. Cooper, U.S. Pat. No. 10,126,058, issued Nov. 13, 2018, to Paul V. Cooper, U.S. Pat. No. 10,126,059, issued Nov. 13, 2018, to Paul V. Cooper, U.S. Pat. No. 10,138,892, issued Nov. 27, 2018, to Paul V. Cooper, U.S. Pat. No. 10,195,664, issued Feb. 5, 2019, to Paul V. Cooper, U.S. Pat. No. 10,267,314, issued Apr. 23, 2019, to Paul V. Cooper, U.S. Pat. No. 10,274,256, issued Apr. 30, 2019, to Paul V. Cooper, U.S. Pat. No. 10,302,361, issued May 28, 2019, to Paul V. Cooper, U.S. Pat. No. 10,309,725, issued Jun. 4, 2019, to Paul V. Cooper, U.S. Pat. No. 10,307,821, issued Jun. 4, 2019, to Paul V. Cooper, U.S. Pat. No. 10,322,451, issued Jun. 18, 2019, to Paul V. Cooper, U.S. Pat. No. 10,345,045, issued Jul. 9, 2019, to Paul V. Cooper, U.S. Pat. No. 10,352,620, issued Jul. 16, 2019, to Paul V. Cooper, U.S. Pat. No. 10,428,821, issued Oct. 1, 2019, to Paul V. Cooper, U.S. Pat. No. 10,458,708, issued Oct. 29, 2019, to Paul V. Cooper, U.S. Pat. No. 10,465,688, issued Nov. 5, 2019, to Paul V. Cooper, U.S. Pat. No. 10,562,097, issued Feb. 18, 2020, to Paul V. Cooper, U.S. Pat. No. 10,570,745, issued Feb. 25, 2020, to Paul V. Cooper, U.S. Pat. No. 10,641,279, issued May 5, 2020, to Paul V. Cooper, U.S. Pat. No. 10,641,270, issued May 5, 2020, to Paul V. Cooper, U.S. Pat. No. 10,675,679, issued Jun. 9, 2020, to Paul V. Cooper, U.S. Pat. No. 10,947,980, issued May 16, 2021, to Paul V. Cooper, U.S. Pat. No. 11,020,798, issued Jun. 1, 2021, to Paul V. Cooper, U.S. Pat. No. 11,098,719, issued Aug. 24, 2021, to Paul V. Cooper, U.S. Pat. No. 11,098,720, issued Aug. 24, 2021, to Paul V. Cooper, U.S. Pat. No. 11,103,920, issued Aug. 31, 2021, to Paul V. Cooper, U.S. Pat. No. 11,130,173, issued Sep. 28, 2021, to Paul V. Cooper, U.S. Pat. No. 11,149,747, issued Oct. 19, 2021, to Paul V. Cooper, U.S. Pat. No. 11,167,345, issued Nov. 9, 2021, to Paul V. Cooper, U.S. Pat. No. 11,185,916, issued Nov. 30, 2021, to Paul V. Cooper, U.S. Pat. No. 11,286,939, issued Mar. 29, 2022, to Paul V. Cooper, U.S. Pat. No. 11,391,293, issued Jul. 19, 2022, to Paul V. Cooper, U.S. Pat. No. 11,471,938, issued Oct. 18, 2022, to Paul V. Cooper, U.S. Pat. No. 11,358,216, issued Jun. 14, 2022, to Paul V. Cooper, U.S. Pat. No. 11,358,217, issued Jun. 14, 2022, to Paul V. Cooper, U.S. Pat. No. 11,519,414, issued Dec. 6, 2022, to Paul V. Cooper, U.S. Pat. No. 11,759,854, issued Sep. 19, 2023, to Paul V. Cooper, U.S. Pat. No. 11,759,853, issued Sep. 19, 2023, to Paul V. Cooper, U.S. patent application Ser. No. 16/413,142, filed May 15, 2019, by Paul V. Cooper, U.S. patent application Ser. No. 16/533,383, filed Aug. 6, 2019, by Paul V. Cooper, U.S. patent application Ser. No. 16/533,404, filed Aug. 6, 2019, by Paul V. Cooper, U.S. patent application Ser. No. 16/877,267, filed May 18, 2020, by Paul V. Cooper, U.S. patent application Ser. No. 16/877,364, filed

May 18, 2020, by Paul V. Cooper, U.S. patent application Ser. No. 17/200,785, filed Mar. 13, 2021, by Paul V. Cooper, U.S. patent application Ser. No. 17/334,259, filed May 28, 2021, by Paul V. Cooper, U.S. patent application Ser. No. 17/496,229, filed Oct. 7, 2021, by Paul V. Cooper, U.S. patent application Ser. No. 17/703,912, filed Mar. 24, 2022, by Paul V. Cooper, U.S. patent application Ser. No. 17/719,274, filed Apr. 12, 2022, by Paul V. Cooper, U.S. patent application Ser. No. 17/826,111, filed May 26, 2022, by Paul V. Cooper, U.S. patent application Ser. No. 17/939,898 filed Sep. 7, 2022, by Vince Fontana, U.S. patent application Ser. No. 18/114,665, filed Feb. 27, 2023, by Paul V. Cooper, U.S. patent application Ser. No. 18/139,936, filed Apr. 26, 2023, by Paul V. Cooper, U.S. patent application Ser. No. 18/480,755, filed Oct. 4, 2023, by Vince Fontana, U.S. patent application Ser. No. 18/502,457, filed Nov. 6, 2023, by Paul V. Cooper.

Three basic types of pumps for pumping molten metal, such as molten aluminum, are utilized: circulation pumps, transfer pumps and gas-release pumps. Circulation pumps are used to circulate the molten metal within a bath, thereby generally equalizing the temperature of the molten metal. Circulation pumps may be used in any vessel, such as in a reverberatory furnace having an external well. The well is usually an extension of the charging well, in which scrap metal is charged (i.e., added).

Standard transfer pumps are generally used to transfer molten metal from one structure to another structure such as a ladle or another furnace. A standard transfer pump has a riser tube connected to a pump discharge and supported by the superstructure. As molten metal is pumped it is pushed up the riser tube (sometimes called a metal-transfer conduit) and out of the riser tube, which generally has an elbow at its upper end, so molten metal is released into a different vessel from which the pump is positioned. Alternate transfer pumping systems can pump molten metal upwards to a launder, which can greatly eliminate turbulence and resulting dross.

Gas-release pumps, such as gas-injection pumps, circulate molten metal while introducing a gas into the molten metal. In the purification of molten metals, particularly aluminum, it is frequently desired to remove dissolved gases such as hydrogen, or dissolved metals, such as magnesium. As is known by those skilled in the art, the removing of dissolved gas is known as "degassing" while the removal of magnesium is known as "demagging." Gas-release pumps may be used for either of both of these purposes or for any other application for which it is desirable to introduce gas into molten metal.

Gas-release pumps generally include a gas-transfer conduit having a first end that is connected to a gas source and a second end submerged in the molten metal bath. Gas is introduced into the first end and is released from the second end into the molten metal. The gas may be released downstream of the pump chamber into either the pump discharge or a metal-transfer conduit extending from the discharge, or into a stream of molten metal exiting either the discharge or the metal-transfer conduit. Alternatively, gas may be released into the pump chamber or upstream of the pump chamber at a position where molten metal enters the pump chamber. The gas may also be released into any suitable location in a molten metal bath.

Molten metal pump casings and rotors often employ a bearing system comprising ceramic rings wherein there are one or more rings on the rotor that align with rings in the pump chamber (such as rings at the inlet and outlet) when the rotor is placed in the pump chamber. The purpose of the

bearing system is to reduce damage to the soft, graphite components, particularly the rotor and pump base, during pump operation.

Generally, a degasser (also called a rotary degasser) includes (1) an impeller shaft having a first end, a second end and a passage for transferring gas, (2) an impeller, and (3) a drive source for rotating the impeller shaft and the impeller. The first end of the impeller shaft is connected to the drive source and to a gas source and the second end is connected to the impeller.

Generally a scrap melter includes an impeller affixed to an end of a drive shaft, and a drive source attached to the other end of the drive shaft for rotating the shaft and the impeller. The movement of the impeller draws molten metal and scrap metal downward into the molten metal bath in order to melt the scrap. A circulation pump is preferably used in conjunction with the scrap melter to circulate the molten metal in order to maintain a relatively constant temperature within the molten metal.

The materials forming the components that contact the molten metal bath should remain relatively stable in the bath. Structural refractory materials, such as graphite or ceramics, that are resistant to disintegration by corrosive attack from the molten metal may be used. As used herein "ceramics" or "ceramic" refers to any oxidized metal (including silicon) or carbon-based material, excluding graphite, or other ceramic material capable of being used in the environment of a molten metal bath. "Graphite" means any type of graphite, whether or not chemically treated. Graphite is particularly suitable for being formed into pump components because it is (a) soft and relatively easy to machine, (b) not as brittle as ceramics and less prone to breakage, and (c) less expensive than ceramics.

Ceramic, however, is more resistant to corrosion by molten aluminum than graphite. It would therefore be advantageous to develop vertical members used in a molten metal device that are comprised of ceramic, but less costly than solid ceramic members, and less prone to breakage than normal ceramic.

SUMMARY

Devices are disclosed that have increased resistance to breakage. One device comprises at least one tension rod positioned inside an outer core. The tension rod and optionally other structures apply tension (or compressive force) to the outer core in order to make it more resistant to breakage. In this disclosure, the tension rod is preferably tightened by in part using a molten metal pump superstructure (also called a platform) that supports the motor. All or most of the outer core is on the side of the superstructure opposite the surface on which the pump is positioned.

The tension rod may be affixed to the outer core by being affixed to a first block of material at the top of the outer core, and affixed to a second block of material at the bottom of the outer core. When the tension rod is tightened, it draws the first block and the second block together which applies axial compressive force to the outer core.

The outer core can be compressed in any suitable manner. If the first block and second block are utilized, the tension rod may be affixed to each by a bolt or other device attached to, and preferably having an area at least about 30% to 150% greater than the cross-sectional area of the tension rod. The bolt or other device could be inside or outside of the first block and/or second block.

A device according to this disclosure, such as a support post or impeller shaft, includes an outer core made of

structural refractory material, such as graphite, graphitized carbon, clay-bonded graphite, carbon-bonded graphite, silicon carbide, ceramics, or the like. The outer core has a first end and a second end and the tension rod includes a first end and a second end. At least one end of the tension rod can extend beyond and terminate outside of the one end of the outer core. Either the first end or the second end of the tension rod, or both, can be tightened against a superstructure. This puts the outer core under compression, and makes the outer core more resistant to breakage. By using the system of the invention, it is also possible to use a thinner cross-sectional outer core wall, thereby reducing material costs.

Also disclosed is a device, such as a support post, for use in molten metal that includes a reinforcement section to strengthen the device and help alleviate breakage.

Also disclosed are molten metal pumps that include one or more devices disclosed herein.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a side, partial cross-sectional view of a support post according to this disclosure.

FIG. 2 is a side, partial cross-sectional view of the support post of FIG. 1 being mounted to a pump superstructure.

FIG. 2B is an optional bottom portion of the support post of FIGS. 1 and 2.

FIG. 2C is a top view of the bottom portion of the support post of FIG. 2B.

FIG. 2D is a cross-sectional view taken along lines D-D of FIG. 2C.

FIG. 2E is a cross-sectional view taken along lines E-E of FIG. 2C.

FIG. 3 is a side view of an alternate support post according to this disclosure.

FIG. 4 is a side, cross-sectional view of the support post of FIG. 3.

FIG. 5 is a top view of the support post of FIG. 3.

FIG. 6 is a partial, side view of the support post of FIG. 3 without the outer casing.

FIG. 7 is a partial, side view of the support post of FIG. 3 without the outer casing.

FIG. 8 is a top view of the support post of FIG. 6.

FIG. 9 is a close up view of detail B of FIG. 7.

FIG. 10 is a side view taken along lines A-A of FIG. 7.

FIG. 11 is a bottom view of the support post of FIGS. 6 and 7.

FIG. 11A is an end view of the support post of FIG. 11.

FIG. 12 is a cross-sectional side view of the support post of FIG. 11 taken along lines E-E.

FIG. 13 is a side view of an alternate support post according to this disclosure.

FIG. 14 is an exploded view of the support post of FIG. 13.

FIG. 15 is a top view of the support post of FIG. 13.

FIG. 16 is a cross-sectional, partial side view of the support post of FIG. 15 taken along lines A-A.

FIG. 17 is a close-up view of detail B shown in FIG. 16.

FIG. 18 is a close-up view of detail C shown in FIG. 16.

FIG. 19 is a side view of the base of the support post of FIGS. 3 and 6.

FIG. 20 is a top view of the base of FIG. 19.

FIG. 21 is a cross-sectional side view taken along line D-D of FIG. 20.

FIG. 22 is a cross-sectional side view taken along line E-E of FIG. 20.

7

FIG. 23 is a perspective, side view of an outer core according to this disclosure.

FIG. 24 is a top view of the outer core of FIG. 23.

FIG. 25 is a side, cross-sectional view of the outer core taken along line F-F of FIG. 24.

FIG. 26 is a perspective side view of a tension rod according to this disclosure.

FIG. 27 is a partial, side view of the tension rod of FIG. 26.

FIG. 28 is a perspective, top view of a support post top according to this disclosure.

FIG. 29 is a top view of the support post top of FIG. 28.

FIG. 30 is a side, cross-sectional view taken along line G-G of FIG. 29.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

For any device described herein, any of the components that contact the molten metal are preferably formed by a material that can withstand the molten metal environment. Preferred materials are oxidation-resistant graphite and ceramic, such as silicon carbide.

FIG. 1 shows a support post 10 in accordance with aspects of the disclosure. Shaft has an outer core 50 that has axial tension applied to it to make outer core 50 more resistant to breakage. Similar techniques, however, may be used to tension rotor shafts or other elongate molten metal pump components. Shaft 10 has a tension rod 20, a top support block 30, a bottom support block 60, an outer core 50, and a bottom 70.

Tension rod 20 is preferably comprised of steel and has a body 24, a first end 24 and a second end 26. As shown, tension rod 20 is threaded along about 5% to 25% of its length starting at first end 24 and moving upward, and along about 10% to 25% of its length starting at second end 26 and moving downward. The threaded portion 24A juxtaposed end 24 is preferably configured to be threaded into a channel 64 in second end 60 and into channel 76A in section 76. Portion 24A need only have sufficient threads to anchor it in second end 60 and/or section 76. Alternatively, shaft 20 need not be threaded into second end 60 and/or section 76, but could instead pass through them and be retained by nut 85 (or other suitable fastener) in section 76 or section 74.

Threaded portion 26A can optionally be threaded partially into bore 39 of top block 30. Nut 40 and nut 120 are threaded onto portion 26A as further described.

Tension rod 20 includes a top, threaded portion 26A that (as shown) threaded partially into top block 30. Top block 30 has an upper portion 34, a top surface 35, an opening 32, a sleeve 38, an internal wall surface 36, and a passage 39. Upper portion 34 is on top of and outside of outer core 50, and surface 36 rests on the top 52 to apply axial tension to outer core 50. Passage 39 is configured so rod 20 can pass therethrough. Opening 32 is formed in top surface 35, is preferably about 1.5 to 2.5 times the diameter of rod 20, and extends into top block 30 from upper surface 35 by about 1" to 3", although opening 32 can be of any suitable dimension. Sleeve 38 fits inside of outer coating 50 and extends downward about 10-30% of the length (although any suitable distance would work, or top block 30 could be stabilized in another manner) of outer coating 50 in order to stabilize top block 30 to outer coating 50.

Channels 80 and 82 are for injecting cement into the bottom of support post 20 to help connect it to a molten metal pump base in a manner known in the art. Any suitable molten metal pump base could be utilized.

8

FIG. 2 shows the support post 10 of FIG. 1 being connected to a superstructure 100 of a molten metal pump, wherein the superstructure 100 supports the pump motor. The superstructure 100 is preferably a steel plate or platform, and is known in the art. Here, it has an opening 102 formed therethrough, a bottom surface 104, and a top surface 106. To add additional tension to outer core 50, a compression spring 110 and nut 120 are positioned on tension rod 20 above surface 106. Nut 120 is then tightened, which ultimately tightens surface 35 of top block 30 against bottom surface 104. Spring 110 need not be used but it or a similar flexible structure is preferred.

Outer core 50 could instead be comprised of graphite and/or blocks 30 and 60 could be comprised of ceramic. Further, any of sections 72, 74, 76 could be comprised of graphite or ceramic.

FIGS. 3-5 show an alternate support post 200 with graphite core 210 and an outer ceramic (preferably silicon carbide) core 250. Alternatively, core 210 could be comprised of ceramic and/or outer core 250 could be comprised of graphite. A reinforcement member 300 is positioned in graphite core 210. In this embodiment outer core 250 is optional. Further, there may be more than one reinforcement member at either one end, or both ends of core 210. Or core 210 could have a single reinforcement member at each end or that extends therethrough or substantially therethrough.

As shown, the reinforcement member 300 is positioned in a manner, and is comprised of a material, such that it helps prevent the core 210 from breaking. Reinforcement member 300 is preferably comprised of steel, has a length of about 10% to 35%, or 15%-25% of the length of core 210, or a length of about 8" to 12", 10" to 16", or 12" to 16", and the cylindrical with a diameter about $\frac{1}{10}$ ", $\frac{1}{8}$ ", $\frac{1}{6}$ ", $\frac{1}{4}$ " or $\frac{1}{2}$ ", or about 10%-30% the diameter of portion 214 of core 210.

Core 210 has a top end 212, a bottom end 214, a top section 212A, a bottom section 214A, and a central portion 216. A bore 220 is formed in core 210 and extends from end 214, preferably through bottom section 214A and partially into section 216. As shown, bore 220 is formed in the center of core 210, although it could be off center.

Reinforcement member 300 is positioned in bore 220 and may be secured by cement. Member 300 has a first end 302 that is preferably tapered and a second end 304. As shown, second end 304 is wider than the body portion 306. A cap 230 is positioned over second end 304 and preferably cemented in place to prevent molten metal from contacting reinforcement member 300. All or part of body portion 306 may be threaded so that member 300 is threaded into bore 220. As shown in FIG. 12, reinforcement member has a smaller-diameter portion 306A that is threaded. Portion 306A is threaded into smaller diameter portion 220A of bore 220. Larger diameter bore portion 220B receives second end 204.

Bores 250 and 252 are for connecting first end 212 of support post 200 to a support post clamp preferably positioned above the superstructure of a molten metal pump.

Some non-limiting examples of the disclosure are as follows:

Example 1: A component for use in a molten metal pump, the component comprising:

- an outer core constructed of graphite or ceramic;
- a tension rod positioned partially inside the outer core, wherein the tension rod has a first end and a second end, and is configured to apply an axial compressive force to the outer core in order to make the outer core less susceptible to breakage;

wherein the first end of the tension rod extends beyond the outer core and has an axially-compressive component positioned thereon, the axially-compressive component positioned against the outer core to place an axial-compressive force on the outer core.

Example 2: The component of example 1, wherein the tension rod has a first end and a second end, the outer core has a first end and a second end, and at least one of the first end or second end of the tension rod extends beyond either the first end or second end of the outer core.

Example 3: The component of example 2, wherein either the first end or the second end of the outer core has a cap, and the end of the tension rod that extends beyond the end of the outer core is tightened against the cap.

Example 4: The component of example 1, wherein the tension rod comprises at least one elongate, metal rod.

Example 5: The component of example 4, wherein the tension rod is comprised of steel.

Example 6: The component of example 1 that is a molten metal pump support post.

Example 7: The component of example 1, wherein the tension rod is secured in the outer core by cement.

Example 8: The component of example 7, wherein the tension rod is bonded to the outer core by the cement.

Example 9: The component of example 1, wherein the outer core comprises graphite.

Example 10: The component of example 1, wherein the outer core comprises silicon carbide.

Example 11: The component of example 1, wherein the outer core comprises material harder than graphite.

Example 12: The component of example 1, wherein the second end of the tension rod is inside of the outer core.

Example 13: The component of example 1, wherein the first end of the tension rod is threaded and the first axially-compressive component is a nut threaded onto the tension rod and tightened against the outer core.

Example 14: The component of example 1 that further includes a second axially-compressive component on the second end of the tension rod.

Example 15: The component of example 1, wherein the second end of the tension rod is threaded and that further comprises a second axially-compressive component at the second end of the tension rod.

Example 16: The component of example 15, wherein the second end of the tension rod is threaded and the second axially-compressive component is a nut threaded into the second end.

Example 17: The component of example 13, wherein the nut is hexagonal.

Example 18: The component of example 16, wherein the nut is hexagonal.

Example 19: The component of example 1 that further comprises a first support block at the first end of the outer core.

Example 20: The component of example 19, wherein the second axially-compressive component is positioned inside of the second support block.

Example 21: The component of example 19, wherein the first support block has a narrow portion positioned inside of the outer core and an enlarged portion that presses against at least part of the wall of the outer core.

Example 22: The component of example 20, wherein the second support block has an extension positioned inside of the outer core and an enlarged portion that presses against at least part of the wall of the outer core to provide axially-compressive force to the outer core.

Example 23: The component of example 1, wherein the second end of the extension rod extends beyond a stationary plate and a third axially-compressive component is positioned on the second end of the extension rod on a side of the stationary plate opposite the outer core, and the third axially-compressive component is compressed to the stationary plate.

Example 24: The component of example 23, wherein the stationary plate is a molten metal pump superstructure.

Example 25: The component of example 23 that includes a compression device between the third axially-compressive component and the stationary plate.

Example 26: The component of example 25, wherein the compression device is a spring.

Example 27: The component of example 19, wherein the first support block is comprised of graphite.

Example 28: The component of example 22, wherein the second support block is comprised of graphite.

Example 29: The component of example 20 that further includes a cap at the second end distal to the second axially-compressive component.

Some other non-limiting examples of the disclosure follow:

Example 1: A support post comprising an elongated body having a longitudinal axis and a height, a first end configured to connect to a superstructure and a second end configured to connect to a pump base, wherein the second end comprises at least one reinforcement section configured to make the second end resistant to breakage.

Example 2: The support post of example 1, wherein the at least one reinforcement section is elongated and has a longitudinal axis.

Example 3: The support post of example 2, wherein the longitudinal axis of the at least one reinforcement section is aligned with the longitudinal axis of the support post.

Example 4: The support post of example 1, wherein the support post is comprised of graphite and the at least one reinforcement section is comprised of one or more of the group consisting of: silicon carbide and steel.

Example 5: The support post of example 1, wherein the at least one reinforcement section is completely surrounded by the material of the support post so the reinforcement section is configured not to contact molten metal.

Example 6: The support post of example 1, wherein the at least one reinforcement section is less than 50% of the height of the support post.

Example 7: The support post of example 1, wherein the at least one reinforcement section is between 15%-35% of the height of the support post.

Example 8: The support post of example 1, wherein the at least one reinforcement section is between 15%-25% of the height of the support post.

Example 9: The support post of example 1, wherein the at least one reinforcement section has a cross-sectional area that is between $\frac{1}{4}$ and $\frac{1}{10}$ the cross-sectional area of the second end of the support post.

Example 10: The support post of example 1, wherein the at least one reinforcement section has a cross-sectional area that is between $\frac{1}{5}$ and $\frac{1}{8}$ the cross-sectional area of the second end of the support post.

Example 11: The support post of example 1, wherein the support post has a bore in its second end and the at least reinforcement section is positioned in the bore.

Example 12: The support post of example 11 that further includes cement in the bore to anchor the at least one reinforcement section.

11

Example 13: The support post of example 1 that further includes a ceramic outer cover.

Example 14: The support post of example 1 that is cylindrical.

Example 15: The support post of example 1, wherein the reinforcement section is cylindrical.

Example 16: The support post of example 1, wherein the second end includes a first portion having a first diameter, and a second portion having a second diameter, wherein the second diameter is less than the first diameter.

Example 17: The support post of example 1, wherein the second end includes a first portion having a first cross-sectional area, and a second portion having a second cross-sectional area is less than the first cross-sectional area.

Example 18: The support post of example 16, wherein the at least one reinforcement section is positioned partially in the first portion and partially in the second portion.

Example 19: The support post of example 17, wherein the reinforcement section is positioned partially in the first portion and partially in the second portion.

Example 20: The support post of example 1 that is cylindrical with a center and the reinforcement section is positioned in the center.

Example 21: The support post of example 1 that further includes one or more channels in the second end, wherein the channels are configured to receive cement.

Example 22: The support post of example 1, wherein the first end is configured to fit into a coupling.

Example 23: The support post of example 11 that further includes a plug at a second tip of the support post, wherein the plug is configured to cover the bore.

Example 24: The support post of example 1 that includes a single reinforcement section.

Example 25: The support post of example 1, wherein the at least one reinforcement section is concrete, positioned in a bore inside of the second end of the support post.

Example 26: The support post of example 1, wherein the at least one reinforcement section extends the length of the support post.

Example 27: The support post of example 1, wherein the at least one reinforcement section has an outer surface including threads, wherein the at least one reinforcement section is threadingly received in the support post.

Example 28: The support post of example 27, wherein the threads are received in the support post at its first diameter and first cross-sectional area.

Example 29: The support post of example 27, wherein the at least one reinforcement section has a length and the threads extend along the entire length.

Example 30: The support post of example 27, wherein the at least one reinforcement section has a length and the threads extend at least 50% of the length.

Example 31: The support post of example 27, wherein the at least one reinforcement section has a length and the threads extend at least 25% of the length.

Example 32: The support post of example 1 that has one or more air-relief grooves.

Example 33: The support post of example 32 that has two air-relief grooves.

Example 34: The support post of example 16, wherein the second diameter is between 3.5" and 4.5".

Example 35: The support post of example 16, wherein the second portion has a height of between 6.0" and 7.0".

Example 36: The support post of example 1, wherein the reinforcement section has a diameter of between 0.75" and 1.25".

12

Having thus described different embodiments, other variations and embodiments that do not depart from the spirit of this disclosure will become apparent to those skilled in the art. The scope of the claims is thus not limited to any particular embodiment, but is instead set forth in the claims and the legal equivalents thereof. Unless expressly stated in the written description or claims, the steps of any method recited in the claims may be performed in any order capable of yielding the desired product. No language in the specification should be construed as indicating that any non-claimed limitation is included in a claim. The terms "a" and "an" in the context of the following claims are to be construed to cover both the singular and the plural, unless otherwise indicated herein.

What is claimed is:

1. A component for use in a molten metal pump, the component comprising:

an inner core and an outer core; and

a tension rod positioned inside the inner core, wherein the tension rod has a first end that is threaded and has a length of 25% or less than a length of the inner core and has a second end that is threaded and has a length that is 25% or less than the length of the inner core, and the tension rod is configured to apply an axial compressive force to the inner core.

2. The component of claim 1, wherein the first end of the tension rod extends beyond the outer core and has an axially-compressive component positioned thereon, the axially-compressive component positioned against the outer core to place an axial-compressive force on the outer core.

3. The component of claim 1, wherein the second end of the tension rod is threaded and that further comprises a third axially-compressive component at the second end of the tension rod.

4. The component of claim 3, wherein the third axially-compressive component is positioned inside of a support block.

5. The component of claim 4, wherein the support block has an extension positioned inside of the outer core and an enlarged portion that presses against at least part of the wall of the outer core to provide axially-compressive force to the outer core.

6. The component of claim 1, wherein the outer core has a first end and a second end, and at least one of the first end or second end of the tension rod extends beyond either the first end or second end of the outer core.

7. The component of claim 6, wherein either the first end or the second end of the outer core has a cap, and each end of the tension rod that extends beyond the end of the outer core is tightened against the cap.

8. The component of claim 1, wherein the tension rod comprises at least one elongate, metal rod.

9. The component of claim 1, wherein the tension rod is comprised of steel.

10. The component of claim 1 that is a molten metal pump support post.

11. The component of claim 1, wherein the outer core comprises graphite.

12. The component of claim 1, wherein the outer core comprises silicon carbide.

13. The component of claim 1, wherein the outer core comprises material harder than graphite.

14. The component of claim 1, wherein the first end of the tension rod is threaded and the first axially-compressive component is a nut threaded onto the tension rod and tightened against the outer core.

15. The component of claim **1**, wherein the inner core has one or more bores that are configured to receive the tension rod.

16. The component of claim **1**, wherein the second end of the extension rod extends beyond a stationary plate and a second axially-compressive component is positioned on the second end of the extension rod on a side of the stationary plate opposite the outer core, and the second axially-compressive component is compressed to the stationary plate.

17. The component of claim **16**, wherein the stationary plate is a molten metal pump superstructure.

18. The component of claim **16** that includes a compression device between the second axially-compressive component and the stationary plate.

19. The component of claim **18**, wherein the compression device is a spring.

20. The component of claim **19** that further includes a cap at the second end distal to the second axially-compressive component.

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